Pandemic Proximity

Did COVID-19 Redraw the Geography of Korea's National R&D Collaboration?

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haseon-jeong.github.io/bio/appam2026

SHORT ANSWER No clear jump in 2020, but distance trends for new partnerships steepened afterward, including for universities and SMEs — a pandemic-era acceleration.

131,813	+0.006 n.s.	+0.073***	≈ +19%	F = 15.9
new partnerships, 2015–2024	2020 departure from the pre-trend	post-2020 change in yearly slope, ln(1 + km)	route distance of new ties, 2024 vs. 2019	pre-trends not flat → not a causal estimate

1 Research Question

Proximity has long been treated as a precondition for R&D collaboration: face-to-face contact, tacit knowledge and trust are easier to build when partners are close.

If remote collaboration became routine during the COVID-19 pandemic, the distance constraint on choosing partners may have loosened. The policy-relevant question is **for whom**.

RQ1 Did partners chosen after 2020 sit farther away?
RQ2 Which institutions and research stages drove the change?
RQ3 What structurally shapes collaboration distance?

2 Data & Design

Data. Korea's National R&D project micro-data (NTIS), 2015–2024: 292,966 tie-year records linking a project's lead institution to each domestic partner, across 65,602 projects. The outcome is the **route (road) distance** in kilometers.

Why new ties? A tie's distance is fixed once the partner is chosen: of 77,200 multi-year ties, only **5.7%** ever change distance. We therefore study **partner selection**. A new tie is a project × partner pair in its first year (history from 2012), giving **131,813 new ties**.

Interrupted time series. Distances were already changing before 2020, so a plain before–after comparison would absorb the old trend. We extend the pre-2020 trend and separate a one-time departure in 2020 (β_2) from a change in the yearly slope (β_3):

$$\ln(1+\text{km})_{ijt} = \beta_1 \text{Year}_t + \beta_2 \text{Post}_t + \beta_3 (\text{Year}_t - 2020)\cdot \text{Post}_t + X_{ijt}\gamma + \mu_{\text{region}} + \varepsilon$$

X: institution type, research stage, ln(project budget), private funding share; region fixed effects. An event study (reference 2019) traces the yearly pattern. Standard errors are clustered by project.

3 Findings

① No clear 2020 jump — the trend steepened afterward

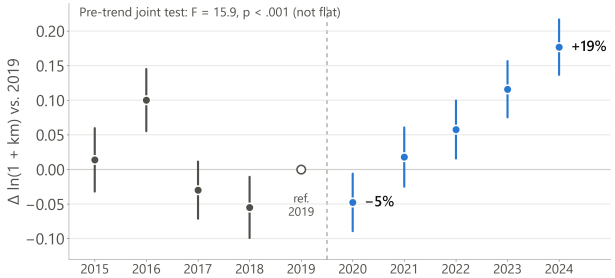


Figure 1. Event study, new ties with controls: ln(1 + km) relative to 2019, 95% CI.

Against the pre-2020 trend, 2020 shows no clear jump (+0.006, p = .76). Afterward the yearly slope turned from −0.018 to **+0.055** (about 5.7% a year), a change of **+0.073 per year**. New partners in 2024 were about **19% farther** than in 2019 (event study). In period means (2015–19 → 2020–24), new ties rose from 122 to 142 km, while continuing ties moved little (138 → 144 km).

The inter-regional share of new ties jumped from 69% to 75% in 2020, but not after controls (−2 pp) — consistent with a shift in *which* projects started.

② Slopes shifted up for universities and SMEs, down for GRIs

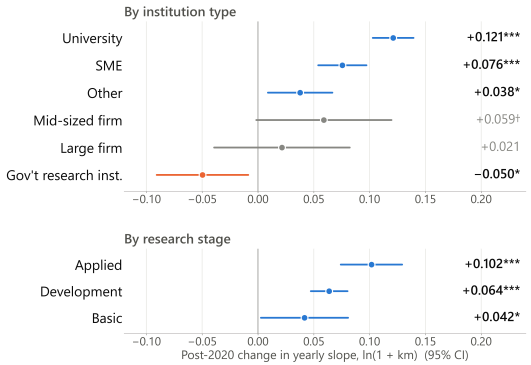


Figure 2. Change in yearly slope after 2020 (not level); one pooled model with group interactions. Blue/orange p < .05; gray n.s.; † p < .1.

The slope rose most for **universities** (+0.121) and **SMEs** (+0.076) — the institutions with the shortest reach before 2020. For government research institutes (GRIs) it shifted down (−0.050): a rising trend flattened. Large firms are **inconclusive** (+0.021, n.s.; ≈ 3,900 ties). Group differences are significant (F = 12.4). By stage, the slope increase was largest for applied research.

Mean km, new ties	2015–19	2020–24
SME	104.7	123.8
University	121.5	152.9
Gov't research institute	173.6	153.8
Large firm	165.6	166.0

Table 1. Short- and long-reach institution types converged.

③ Gaps persist across institutions and stages

In level terms (OLS with year FE, N = 130,890), **SMEs remain the most local** — about 33% shorter than large firms, holding other factors fixed — and distance falls from basic to development research. Larger budgets go with farther partners; private co-funding with closer ones (associations, not effects).

4 Robustness & Limitations

Post-2020 slope change	β_3 / yr
New ties, controls (main)	**+0.073*** **
No controls	+0.029***
Official new-project flag (excl. 2020)	+0.069***
Excluding 2024	+0.073***
Quadratic time trend	+0.065*
Break year 2021	+0.071***

Pre-trends are not flat (F = 15.9) and placebo breaks in 2017 and 2018 are also significant, so this is a *pandemic-era acceleration* — a change observed in that period, not a causal effect. 2022–24 also overlaps other R&D policy changes (e.g., the 2024 budget restructuring).

5 Takeaways

1. **Distance gaps across institution types narrowed.** Whether this widened *access* is the next question.
2. **Gaps persist.** SMEs and development-stage projects remain the most local.
3. **Policy options to evaluate:** matching intermediaries for SMEs, minimum-scale consortia, stage-differentiated incentives for development projects.

Data: NTIS National R&D project micro-data, 2015–2024. *** p < .001, ** p < .01, * p < .05, † p < .1. Also positive: all tie-year records +0.052***; inter-regional tie share (LPM) +0.022***.